3. n = 4

l = 0, 1, 2, 3

∴ 4 sub shells s, p, d & f.

l = 0 m_l = 0 ⇒ one 4s orbital.

l = 1 m_l = −1, 0, + 1 ⇒ three 4p orbitals.

l = 2 m_l = −2, −1, 0, +1, +2 ⇒ five 4d orbitals.

l = 3 m_l = −3, −2, −1, 0, +1, +2, +3 ⇒ seven 4f orbitals.

Over all 16 orbitals are possible.

4.

Orbital	**n**	***l***	**Radial node n – *l* – 1**	**Angular node *l***	**Total node n – 1**
3d	3	2	0	2	2
4f	4	3	0	3	3

5. $E_n = \frac{-13.6}{n^2}$ eV

Second excited state

$n = 3 \quad \therefore E_3 = \frac{-13.6}{9}$ eV

$E_3 = -1.51$ eV

6. Electronic configuration of Fe^{3+} is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^0\ 3d^5$

Five unpaired electrons.

1	1	1	1	1

$3d^5$

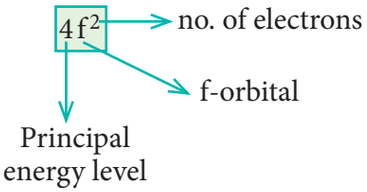
Electronic configuration of Mn^{2+} is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^0\ 3d^5$

∴ **Five unpaired electrons.**

Electronic configuration of Ar: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$

no unpaired electrons.

7.



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